

Nature-Inspired AI — Exam (2026-06-22)

reconstructed from transcript, with answer key

The exam is shown unmarked; correct answers and justifications are in the answer key at the end.

Exam

1. A smart agriculture company wants to use GP to evolve programs that control an autonomous irrigation drone. The drone operates over a rectangular crop field divided into a 10×10 grid of cells. At each time step the drone senses its environment and must decide what action to take.
Sensors: `soil_moisture(x,y)` — moisture of cell (x,y) , float in $[0,1]$; `crop_health(x,y)` — health index of cell (x,y) , float in $[0,1]$; `water_remaining` — water left, float in $[0,1]$; `current_x`, `current_y` — drone's position.
Atomic actions: `move_north()`, `move_south()`, `move_east()`, `move_west()` (move one cell); `irrigate()` (water the current cell, consuming 0.1 units of water); `skip()` (do nothing this step).
Fitness = number of cells with `soil_moisture` > 0.6 at the end of the episode.
A GP expert raises several concerns. The designer responds with four possible redesigns; only one correctly fixes *all* the flaws in both the primitive set and the fitness function. Which is it?
 - a) The fitness function is flawed because it does not account for water efficiency or crop health, and could be gamed by a program that simply irrigates every cell regardless of actual need. The primitive set is missing control-flow functions such as `if_then_else` and `seq` in the Function Set, without which the GP tree cannot express conditional irrigation logic. A corrected design adds `if_then_else(condition, a, b)` and `seq(a, b)` to the Function Set, and replaces the fitness with one that combines the number of improved crop-health cells, a penalty for water wasted on already-moist cells, and a penalty for low-health cells that were never visited.
 - b) The fitness function is correct and complete. The primitive set should remove `soil_moisture(x,y)` and `crop_health(x,y)` from the Terminal Set because floating-point sensors violate the closure property, and only Boolean sensors should be used in GP.
 - c) The only flaw is that `water_remaining` should not be in the Terminal Set because it changes over time, making the tree non-deterministic. The fitness should be replaced with the total number of `irrigate()` calls made during the episode.
 - d) The primitive set is fine as-is. The fitness function should be replaced with the total distance travelled by the drone to encourage efficient movement.
2. What is the standard recombination operator in GP?
 - a) Uniform crossover
 - b) One-point crossover on bit string
 - c) Exchange subtrees
 - d) Arithmetic recombination

3. In DE survival selection, when does the trial vector u_i replace the target vector x_i ?
 - a) Whenever the trial vector passes the crossover probability threshold
 - b) When a uniform random condition is satisfied
 - c) When $f(u_i) < f(x_i)$, i.e. the trial is worse (for min)
 - d) When $f(u_i) > f(x_i)$, i.e. the trial is better than the target
4. In arithmetic crossover for real-valued parents x_1 and x_2 with $\alpha \in [0, 1]$, the offspring x' is:
 - a) $x' = x_1 + x_2$
 - b) $x' = \alpha \times (x_2 - x_1)$
 - c) $x' = \alpha x_1 + (1 - \alpha) x_2$
 - d) $x' = (x_1 \times x_2) / \alpha$
5. Which is an advantage of EAs over traditional gradient-based optimization methods?
 - a) EAs do not require gradient information and can handle noisy, black-box or non-differentiable objective functions
 - b) EAs are guaranteed to find the global optimum in polynomial time
 - c) EAs are always faster for any problem
 - d) EAs require a small population when the search space is high-dimensional
6. In ACO, pheromone evaporation serves the purpose:
 - a) It is used only in the Rank-Based Ant System
 - b) It replaces the cognitive component from PSO
 - c) It prevents unlimited accumulation of pheromone and allows the algorithm to forget poor solutions over time
 - d) It increases pheromone on the best path found so far
7. Which of the following is NOT a type of terminal?
 - a) Variables (x, y, z)
 - b) Zero-arity functions (rand, go-left)
 - c) Conditional operators (if-then-else)
 - d) Constant values $(0, 3.5)$
8. NES differs from CMA-ES and Simple ES because it:
 - a) Uses all sampled solutions, including poor ones, to update the probability distribution via the natural gradient
 - b) Replaces mutation with crossover
 - c) Only keeps the best solution
 - d) Does not use a probability distribution to sample solutions
9. You are using GP to evolve sorting algorithms for integer arrays of size n . Your **Function Set** includes `seq(a, b)` (execute a then b), `swap(i, j)` (swap elements at positions i and j), `if_greater(i, j, a, b)` (if element at i is greater than element at j , execute a, else execute b), and `repeat(k, a)` (repeat instruction a exactly k times). A colleague proposes the following **Terminal Set**: $\{i, j, 0, 1, n-1, +, -, \text{rand}()\}$. Which of the following correctly identifies the flaws in this Terminal Set?

- a) It is problematic because $+$ and $-$ are functions that take arguments, not zero-arity terminals, and `rand()` introduces non-determinism that makes fitness evaluation inconsistent; a cleaner Terminal Set would be $\{i, j, 0, 1, n-1\}$ with arithmetic operators moved to the Function Set if needed.
 - b) It is incorrect because terminals should only ever contain Boolean values when the Function Set includes conditionals.
 - c) It is entirely correct — all those terminals are necessary and sufficient for evolving sorting algorithms.
 - d) It is mostly correct but `rand()` should be removed, as non-deterministic terminals make fitness evaluation inconsistent across runs.
10. Which population initialization will always fill trees with valid functions until the maximum depth?
- a) Ramped half-and-half
 - b) Random
 - c) Full
 - d) Grow
11. Which of the following describes the role of temperature in SA?
- a) Determines the number of neighbors evaluated
 - b) Defines the size of the initial population
 - c) Sets the maximum number of iterations
 - d) Controls the probability of accepting a worse solution, enabling exploration when high and exploitation when low
12. Apply OX to the parents, keeping positions 2 to 5 (0-indexed) from x_1 :
- $$x_1 = [2, 4, 7, 1, 6, 3, 8, 5, 9], \quad x_2 = [5, 3, 1, 9, 6, 2, 4, 8, 7]$$
- a) $[9, 2, 7, 1, 6, 3, 4, 8, 5]$
 - b) $[5, 9, 7, 1, 6, 3, 4, 2, 9]$
 - c) $[2, 4, 7, 1, 6, 3, 8, 5, 9]$
 - d) $[9, 2, 7, 1, 6, 3, 8, 4, 5]$
13. Using the best individual as the base vector in DE:
- a) Eliminates the need for crossover
 - b) Increases selective pressure and decreases diversity
 - c) Increases diversity, decreases selective pressure
 - d) Has no effect on selection or diversity
14. In GP, the result classifies the training data perfectly but fails on unseen data. Explanation and remedy:
- a) Population too small; increase it
 - b) The program has overfit; use a validation set and penalize complexity
 - c) Mutation rate too high; reduce it
 - d) Crossover rate too low; increase it

15. In $(\mu + \lambda)$ -ES, which individuals are considered in selection?

- a) Only the λ offspring
- b) The μ parents
- c) A random subset of λ and μ
- d) Both μ and λ

16. In PSO, the cognitive velocity is:

- a) $k \times (x_{\text{best}} - x^{(j-1)})$
- b) $k \times v^{(j-1)}$
- c) $k \times (x_{\text{global-best}} - x^{(j-1)})$
- d) $k \times (x^{(j)} - x_{\text{best}})$

Answer Key

Every *circled* answer on the original sheet (Q2–Q16) matches the key. ✓ marks the correct option below.

- Q1** ✓ **a.** Only (a) fixes both flaws. The fitness merely counts wet cells, so it is gamed by a program that irrigates everywhere — it ignores water efficiency and `crop_health`; and the Function Set lacks control flow (`if-then-else`, `seq`), so the tree cannot express conditional irrigation. (b) wrongly calls float sensors a closure violation; (c) misdiagnoses `water_remaining` and proposes a gameable fitness; (d) ignores the fitness flaw.
- Q2** ✓ **c.** Subtree exchange is the canonical GP crossover.
- Q3** ✓ **d.** DE keeps the better of {target, trial} (slide: “choose the better solution”). The item’s inequalities and “better/worse” labels are inconsistent on min vs. max; **d** is the intended “trial is better”.
- Q4** ✓ **c.** Whole-arithmetic crossover = convex combination.
- Q5** ✓ **a.** b/c/d are false absolutes (no guarantees, no free lunch).
- Q6** ✓ **c.** Evaporation = decay preventing stagnation; (d) describes deposit.
- Q7** ✓ **c.** `if-then-else` takes arguments, so it is a function (internal node), not a terminal.
- Q8** ✓ **a.** NES = natural-gradient update on the search distribution using all samples.
- Q9** ✓ **a.** `+` and `-` are binary functions (arity 2), not zero-arity terminals — a leaf cannot take arguments — so they belong in the Function Set; and a runtime `rand()` makes fitness evaluation non-deterministic. (b) is false (no Boolean-only rule); (c) wrongly calls the set correct as-is; (d) catches only `rand()` and misses the `+/-` arity error.
- Q10** ✓ **c.** Full uses only functions until the depth limit, then terminals; Grow may stop early.
- Q11** ✓ **d.** High $T \rightarrow$ explore (accept worse), low $T \rightarrow$ exploit.
- Q12** ✓ **a.** Working:

Q12 — OX working

- Copy the block.** 0-indexed positions 2–5 of x_1 are [7, 1, 6, 3], placed at the same slots:
child = [_, _, **7**, **1**, **6**, **3**, _, _, _].
- Fill from x_2 ,** starting just after the block (position 7), wrapping, skipping values already present {7, 1, 6, 3}:
 x_2 from pos 7: 4, 8, ~~7~~, 5, ~~3~~, ~~1~~, 9, ~~6~~, 2 $\Rightarrow$ 4, 8, 5, 9, 2.
- Drop into the empty slots** (7,8,9 then wrap to 1,2):
pos7= 4, pos8= 8, pos9= 5, pos1= 9, pos2= 2.
 $\Rightarrow$ [9, 2, 7, 1, 6, 3, 4, 8, 5] = option **a**.
(Option d swaps the 4 and 8 — the trap if you fill x_2 from its start instead of from after the cut.)

- Q13** ✓ **b.** Confirmed by DE slide: “ $v_i = x_{\text{best}} + \dots \rightarrow$ increase selective pressure, decrease diversity.”
- Q14** ✓ **b.** Perfect-train / poor-test is the signature of overfitting; remedy with validation + complexity penalty.
- Q15** ✓ **d.** “+” means parents and offspring compete together (elitist); “,” would be offspring only.
- Q16** ✓ **a.** Cognitive term pulls toward the particle’s own personal best; (c) is the social term, (b) is inertia.